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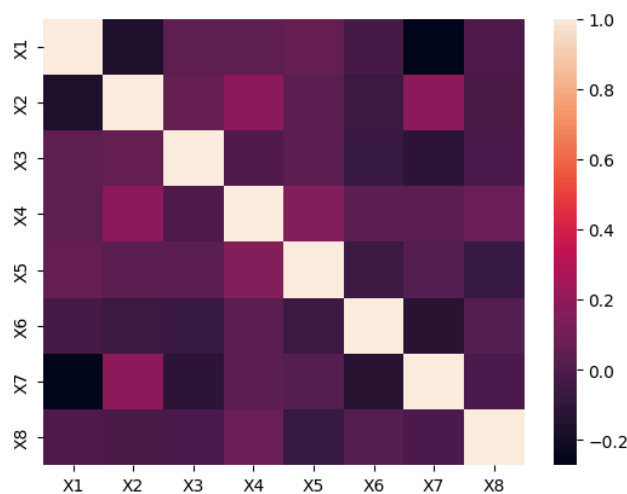
Assignment 01 Technical Report

The objective of this assignment is to analyze and determine the results of three models: Support Vector Machine, Logistic Regression, and k-Nearest Neighbors on a numeric dataset. Each model was tasked to use multiple predictor variables to classify target variable "Y." Each model's performance was evaluated based on accuracy, confusion matrices, and ROC/AUC scores, determining the model that best fits for use in this case of binary classification.

Analysis

Before the models were utilized, [the given dataset](#) was analyzed for any possible multicollinearity between variables. This was achieved through creating the correlation heatmap below.

Correlation between X Variables



All of the "X" variables showed low to moderate correlations, and none indicated any significant multicollinearity issues, which could impact the interpretability of some models like logistic regression. In this case, the data seems to have healthy variation among each other. The target variable, "Y," is binary with labels "A" and "B," serving as the classes to be predicted by each of the three models.

Methodology

After preprocessing and z-scoring of the given dataset, three distinct models were built in the following order: SVM, Logistic Regression, and KNN, using cross-validation for parameter selection.

The SVM model finds the optimal hyperplane separating two classes, which is done by maximizing the distance between the hyperplane and the nearest data points from each class. Here, the hyperparameters were tuned using GridSearchCV to find the best ones based on cross-validation performance. Three parameters were identified:

1. Kernel: This function defines the input space to enhance class separability. Two options were considered: `linear` and `rbf`, good for linear and nonlinear relationships, respectively. The grid search identified `rbf` as the optimal kernel, suggesting that the data benefits from a more flexible boundary.
2. Regularization Parameter: Controls the trade-off between classifying training data and maintaining a smooth decision boundary. A larger value allows the model to fit the data more closely. The grid search selected `50` as the optimal value, indicating a model that is not overly regularized.
3. Gamma: Specifically for the RBF kernel, gamma determines the influence of individual data points on the decision boundary. A low gamma considers distant data points, while a high gamma creates more complex boundaries. The grid search chose 0.01 as the best value.

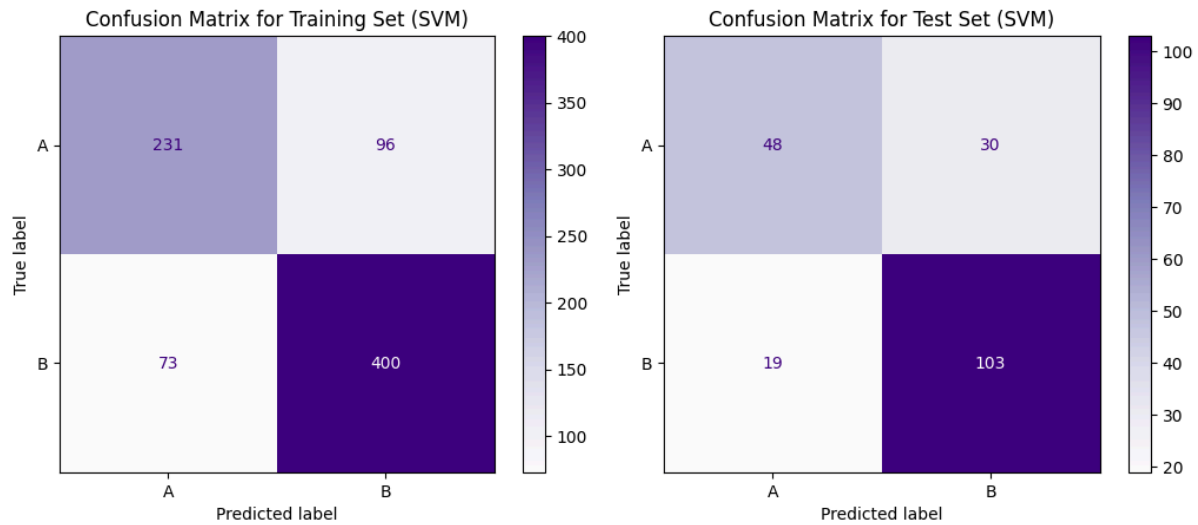
The final SVM model was then evaluated on its train and test sets.

Subsequently, a logistic regression model was constructed. This model, which fits a linear decision boundary between two binary classes, does not require much parameter tuning. Lastly, a K-Nearest Neighbors model was built. KNN classifies new data points based on the majority class of the nearest neighbors in a space. A grid search was used to select the best `n\_neighbors` parameter, ultimately choosing `1`, which suggests that each point is classified based on its single closest neighbor.

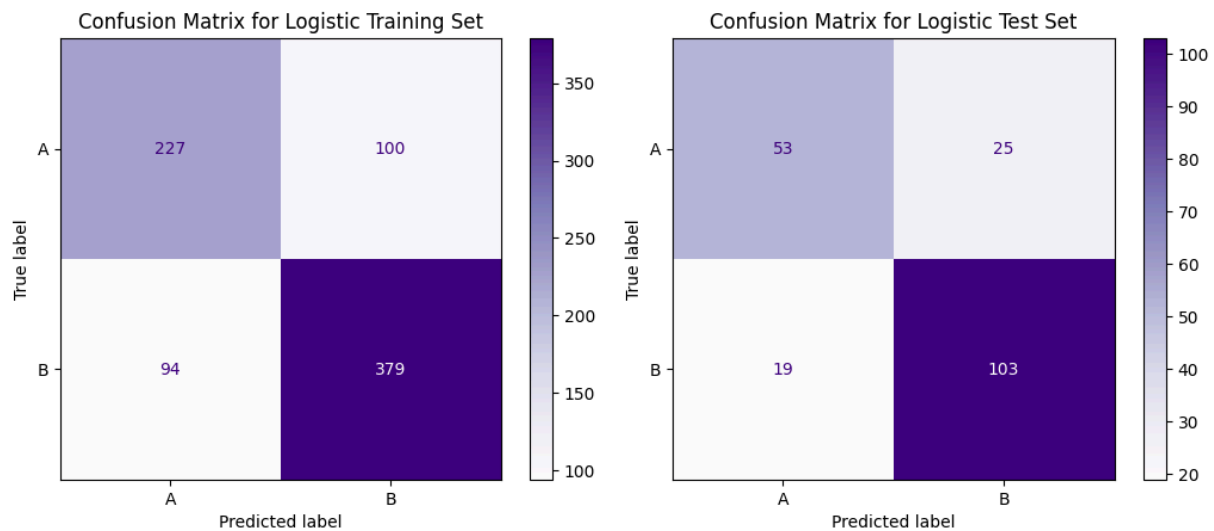
Results

The models' performance was measured by multiple methods: Accuracy Score, ROC/AUC curves, and confusion matrices for training and testing sets. The ROC/AUC scores as well as the confusion matrices will be the most insightful, as they are the most specific in showing how many predictions the models made were true.

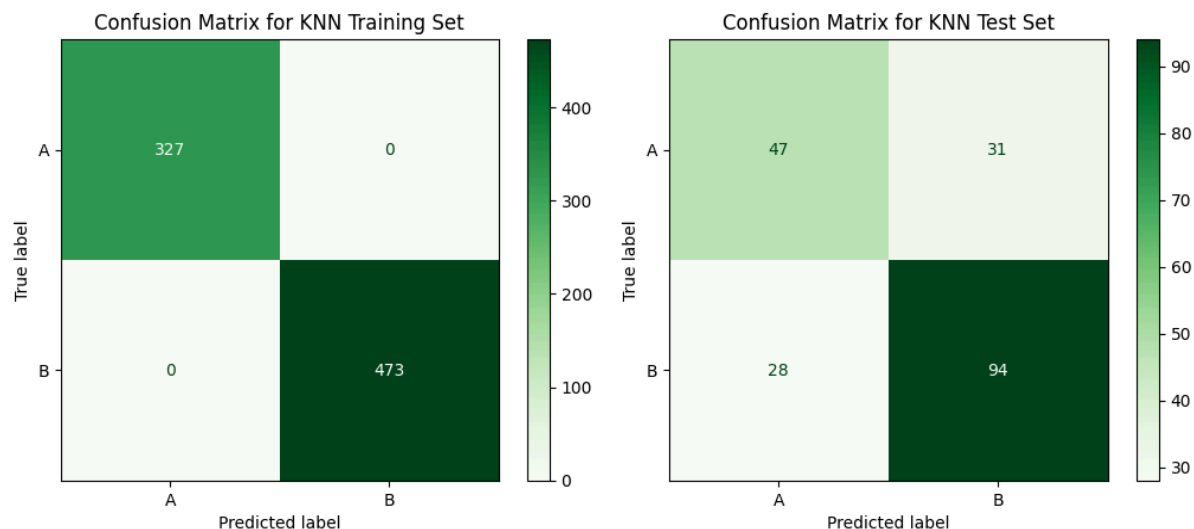
The results for the SVM included a Train Accuracy of 0.78875, a Test Accuracy of 0.755, a Train ROC/AUC of 0.858, and a Test ROC/AUC of 0.835. The confusion matrices for training and testing are shown below:



Logistic regression performed relatively similarly, with a Train Accuracy of 0.7575, Test Accuracy of 0.78, a Train ROC/AUC of 0.8395, and a Test ROC/AUC of 0.8668. The confusion matrices for logistic regression show that this model was also relatively balanced, with a slight edge over SVM in test accuracy.



The KNN model, while achieving a perfect train accuracy, had a much lower test accuracy of 0.705. The train ROC/AUC score was also perfect, but the test score dropped to 0.686, indicating significant overfitting. The confusion matrices for KNN clearly show this as well, with the model failing to generalize beyond the training data.



Overall, both SVM and logistic regression displayed balanced performance, with logistic regression slightly outperforming SVM in accuracy. The KNN model, on the other hand, struggled to generalize, as evidenced by its overfitting.

Model Comparison & Recommendation

The decision of which model to deploy depends between interpretability and flexibility. Given the comparable test accuracies and AUC/ROC scores, logistic regression is recommended due to its simplicity. However, if the data were to show more complex relationships that cannot be captured linearly, the SVM model is also preferable. The KNN model is not recommended, however, it is worth exploring more with different parameters.

Reflection

This analysis painted a picture of the importance of hyperparameter tuning and cross-validation in building different models. The computational expense of grid search was a challenge, particularly for SVM and KNN, but it was necessary to find the best balance of parameters. The KNN model's overfitting highlighted the need for careful parameter selection to prevent the model from learning noise in the training data. In future analyses, additional techniques could enhance performance. In conclusion, this project provided a deeper understanding of the trade-offs between model flexibility, interpretability, and generalizability in machine learning.